

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Data Science	Course Code:	CS4048
	Program:	BS (CS) and BS (SE)	Semester:	Spring 2026
	Duration:	180 Minutes	Total Marks:	100
	Paper Date:	May 21, 2026	Weight	40 %
	Section:	All Sections	Page(s):	4
	Exam Type:	Final Exam		

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 Instruction/Notes: Write concise answers.

Q1. (20 Marks) CLO: 2

Consider the following transactional dataset of a retail grocery shop, where each transaction represents items purchased together by a customer during a single visit:

TID	Items
1	F lour, A pple, C offee, D ates, G hee, I nsant N oodles, M ilk, P eanut B utter
2	A pple, B read, C offee, F lour, L entils, M ilk, O live O il
3	B read, F lour, H oney, J am, O live O il
4	B read, C offee, K itchen T owels, S oap, P eanut B utter
5	A pple, F lour, C offee, E ggs, L entils, P eanut B utter, M ilk, N ut C utter

Total = 33

Assume the **minimum support threshold** = 3.

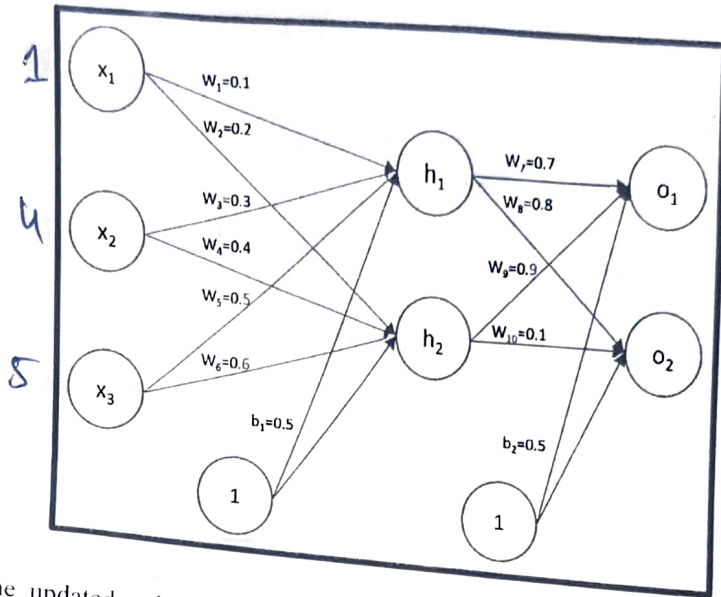
Apply the **FP-Growth algorithm** on the given dataset.

Clearly show all calculations including:

1. Frequency of items
2. Construction of the FP-Tree
3. Conditional pattern bases
4. Conditional FP-Trees
5. **most frequent pattern(s)** in the dataset.

Properly draw the **FP-Tree** and show each step of the mining process.

Q2. (20 marks) CLO: 3



Calculate the updated values of weights w_1 , w_3 , w_7 , and w_9 using the Backpropagation Algorithm for the given multi-layer feed-forward neural network.

Given

Inputs: $x_1=1$, $x_2=4$, $x_3=5$

Targets: t_1 at $o_1 = 0.1$, t_2 at $o_2 = 0.05$

Learning rate: $\alpha=0.01$

Using the initial weights provided in the diagram:

1. Perform the forward pass using the sigmoid activation function.
2. Calculate the error using Mean Squared Error (MSE).
3. Apply backpropagation to compute gradients.
4. Update weights w_1 , w_3 , w_7 , and w_9 after one iteration.

Q3. (20 marks) CLO: 2 Given the following two data points in a two-dimensional feature space:

$$x_1 = [1, 1.5, 5, 3, 4, 3]$$

$$x_2 = [1, 1.5, 5, 4, 4, 3.5]$$

Apply **Agglomerative Hierarchical Clustering** using the **single linkage criterion** to group the data points. Construct the corresponding dendrogram and calculate the lifetime of each cluster.

Q4. (20 marks) CLO: 2

Given the following two-dimensional (2D) data points, apply the Principal Component Analysis technique to reduce the dimensionality of the dataset from 2D to 1D.

Perform the following steps in sequence:

1. Center the dataset by subtracting the mean from each feature.
2. Compute the covariance matrix of the centered data.
3. Determine all principal components (eigenvectors) and their corresponding eigenvalues.
4. Normalize the eigenvectors and identify the principal component used for dimensionality reduction.
5. Project (construct) the data from 2D to 1D using the selected principal component.
6. Reconstruct the data back to the original 2D space from the reduced representation.
7. Evaluate the reconstruction quality by calculating the Mean Squared Error (MSE) between the original and reconstructed data to analyze the information loss resulting from dimensionality reduction.

X1	X2
2	0
0	2
3	1
4	3

Q5. (20 marks) CLO: 3

- Suppose "Risk" is the class label.
- (a) Calculate the entropy of "Risk". Show the calculation steps.
- (b) Calculate the average conditional entropy of "Continent" and then calculate the Information Gain if it is to be used as the root node. Show the calculation steps.
- (c) Suppose that the attribute "Scaled" has been chosen as the root node. It has two branches "yes" and "no". Assume that the attribute "Continent" has been chosen as the next node on the branch "yes". Which are the attributes whose entropies will have to be calculated to find the most suitable node for the branch "no".

yes / no

No.	Continent	Scaled	Death rate	Risk	Height
1.	Asia	yes	high	high	high
2.	Africa	yes	high	high	medium
3.	Africa	yes	low	none	medium
4.	Europe	yes	low	none	low
5.	Asia	no	low	medium	high
6.	Asia	no	low	medium	high
7.	Europe	no	low	none	low
8.	Asia	yes	low	none	medium
9.	Asia	no	low	none	high
10.	Europe	yes	high	high	high
11.	Africa	yes	high	high	low
12.	Asia	yes	high	high	medium
13.	Africa	yes	high	high	high
14.	Africa	yes	low	medium	medium
15.	Africa	yes	low	medium	high
16.	Africa	yes	low	none	low
17.	America	yes	low	none	low
18.	America	yes	low	medium	high
19.	America	yes	low	none	low
20.	America	yes	low	medium	high
				none	low

